Paper 9 - Hamiltonian Window Emergence

CQE-paper-09

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

Describe local inside global temporal emergence as Hamiltonian-window readout over transported states.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-09.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-09\paper_kernel_manifest.json
- body: production\papers\CQE-paper-09\PAPER-BODY.md
- source: production\papers\CQE-paper-09\SOURCE.md
- formal: production\papers\CQE-paper-09\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-09\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-09\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-09.md

Paper 9 - Hamiltonian Window Emergence

Abstract

Paper 9 begins the second block of the CQECMPLX suite by proving finite Hamiltonian window emergence over carried centers. Paper 8 supplies the closure frame from the first block. Paper 9 reads ordered local centers through odd-width windows and emits composite centers with explicit forward and backward receipts.

The production instance uses widths:

```
3, 5, 7
```

Starting with six base center states, these reads emit:

```
width 3 -> 4 surviving centers
width 5 -> 3 surviving centers
width 7 -> 2 surviving centers
```

The theorem proves finite window arithmetic and provenance preservation. It does not prove physical time reversal, static-Z4 temporal periodicity, or higher-order convergence.

Claims

Claim 9.1. A Hamiltonian window of width `w` over an ordered sequence of `n` center states emits exactly `n-w+1` composite centers.

Claim 9.2. Each emitted composite center preserves source indices, source papers, source centers, forward receipt, and backward receipt.

Claim 9.3. In the production sequence, width-3, width-5, and width-7 reads emit exactly four, three, and two composite centers.

Claim 9.4. A reverse receipt is receipt-level reversibility, not proof of physical time reversal.

Definitions

A **center state** is:

```
(paper_id, center)
```

A **Hamiltonian window** is a contiguous slice of fixed odd width `w` over an ordered center-state sequence.

A **forward receipt** records the center sequence in source order:

```
C_i -> C_{i+1} -> ... -> C_{i+w-1}
```

A **backward receipt** records the same centers in reverse order:

```
C_{i+w-1} <- ... <- C_{i+1} <- C_i
```

A **surviving composite center** is:

```
C[C_i|C_{i+1}|...|C_{i+w-1}]
```

accepted only when the source indices and centers are preserved.

Theorem 9.1 - Hamiltonian Window Emergence

For any finite ordered sequence of center states and any fixed window width `w <= n`, the Hamiltonian window read emits exactly `n-w+1` surviving composite centers. Each composite center preserves its source indices, source centers, forward receipt, and backward receipt.

Proof

Let:

```
S = [s_0, s_1, ..., s_{n-1}]
```

be a finite ordered sequence. For width `w <= n`, the valid starts are:

```
0, 1, ..., n-w
```

There are `n-w+1` valid starts and therefore `n-w+1` windows.

For each start index `i`, define:

```
W_i = [s_i, s_{i+1}, ..., s_{i+w-1}]
```

The forward receipt records centers in the order of `W_i`. The backward receipt records the same centers in reverse order. Reversing the backward receipt recovers the forward center sequence, so the source-center provenance is preserved. The composite center is the ordered join of those same source centers.

For the production instance, begin with six base centers. Width `3` yields:

```
6 - 3 + 1 = 4
```

Appending the order-2 state gives seven states. Width `5` yields:

```
7 - 5 + 1 = 3
```

Appending the order-3 state gives eight states. Width `7` yields:

```
8 - 7 + 1 = 2
```

This proves the reported counts and the provenance-preserving construction.

Receipt

The primary executable receipt is:

```
production/formal-papers/CQE-paper-09/verify_hamiltonian_window_emergence.py
production/formal-papers/CQE-paper-09/hamiltonian_window_emergence_receipt.json
```

The receipt status is `pass`. It verifies:

```
order2_width3_count_is_four                    = true
order3_width5_count_is_three                   = true
order4_width7_count_is_two                     = true
all_reads_preserve_source_indices              = true
all_reads_preserve_source_centers              = true
all_backward_receipts_reverse_to_forward       = true
static_z4_does_not_prove_temporal_periodicity = true
```

Falsifiers

The verifier rejects:

```
the static Z4 chart template proves the temporal trace is period 4
a reverse receipt proves physical time reversal
a composite center is valid without source-window provenance
```

These are overclaims relative to the finite window theorem.

Role in the Suite

Paper 9 turns the first-block closure scaffold into ordered center-window construction:

```
verified closure scaffold
-> ordered carried centers
-> width-3, width-5, width-7 reads
-> composite temporal states with receipts
```

This is the suite's first controlled temporal-emergence rule. It is controlled because every composite center must carry its source provenance.

Open Audit Boundaries

- Reverse readability is not physical time reversal.
- Static Z4 chart symmetry does not force Rule 30 temporal periodicity.
- Higher-order convergence beyond width 7 is not claimed here.
- Physical Hamiltonian dynamics assigned to composite centers require a later theorem.

Conclusion

Paper 9 proves temporal construction as finite local-window emergence. It shows how carried centers become composite centers with replayable receipts while refusing to confuse receipt reversal or static labels with physical time dynamics.